Trainings

Initial Check

This procedure covers both rear gear train engines and front gear train engines.

Prior to starting this procedure, make sure there is adequate clearance to remove the camshaft from the rear of the engine for rear gear train engines and from the front of the engine for front gear train engines.

Clearance from Front/Rear Gear Housing			
	mm		in
4 Cylinder	60.96	MIN	24
6 Cylinder	81.28	MIN	32

Note : It may be necessary to remove original equipment manufacturer (OEM) components (radiator, charge-air cooler assembly, etc.) for access. Refer to the OEM instructions.

If adequate clearance can **not** be obtained, the engine **must** be removed.

Preparatory Steps

Rear Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Support the rear of the engine using the rear support attached to the rear of the cylinder block. Failure to support the engine can cause personal injury.

Note : The camshaft must be removed from the flywheel end of the engine.

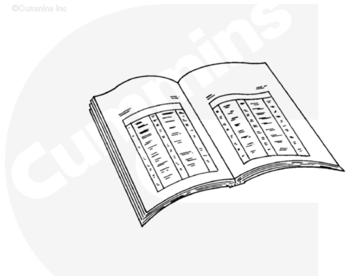
- Disconnect the batteries.
- Remove the transmission and all related components, if equipped. Refer to the OEM service manual.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html)
- Remove the flexplate. Refer to Procedure 016-004 in Section 16. (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16 (/qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html)
- Remove the fuel lift pump. Refer to Procedure 005-045 in Section 5 (/qs3/pubsys2/xml/en/procedures/40/40-005-045.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Remove the rocker levers. Refer to Procedure 003-008 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4 (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- Raise the tappets. Refer to Procedure 004-015 in Section 4. (/qs3/pubsys2/xml/en/procedures/40/40-004-015-tr.html)
- Lock the fuel pump. Refer to Procedure 005-014 in Section 5 (/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html).

Note : Failure to lock the fuel pump may result in improper fuel pump timing during reassembly.

Front Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Note : The camshaft must be removed from the vibration damper end of the engine.

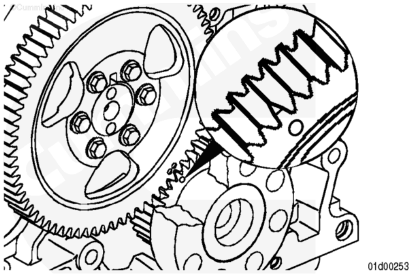
- Disconnect the batteries.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Remove the rocker levers. Refer to Procedure 003-008 in Section 3. (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- Remove the fuel lift pump. Refer to Procedure 005-045 in Section 5. (/qs3/pubsys2/xml/en/procedures/40/40-005-045.html)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- Remove the fan hub, if required. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
- Remove the vibration damper. Use the following procedure, if equipped with a viscous damper. Refer to Procedure 001-052 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html)
- Remove the vibration damper. Use the following procedure, if equipped with a rubber damper. Refer to Procedure 016-004 in Section 16. (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html)
- Raise the tappets. Refer to Procedure 004-015 in Section 4 (/qs3/pubsys2/xml/en/procedures/40/40-004-015-

tr.html)

Remove

Rear Gear Train

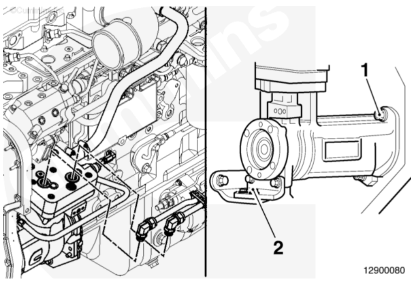
Note : The engine can have either a mark on the crankshaft gear or a chamfered tooth.



Rotate the engine to align the timing marks on the camshaft and crankshaft gear.

Service Tip: The engine can be rotated by installing two of the flywheel/flexplate mounting capscrews half way. Then use a pry bar in between the two capscrews to rotate the engine.

Service Tip: Engines equipped with air compressors may require the air compressor be timed to the engine. To make sure that the air compressor is properly timed when the camshaft gear is later installed, scribe an alignment line on the air compressor and camshaft gear before removing the camshaft gear.



On engines equipped with an air compressor/accessory drive, it may be necessary to loosen/remove some of the air compressor/accessory drive mounting hardware in order to remove the camshaft gear. It is **not** necessary to remove the air compressor/accessory drive completely.

Loosening/removing some of the air compressor/accessory drive mounting hardware will give enough clearance to remove the camshaft gear.

Loosen the air compressor mounting fasteners (1).

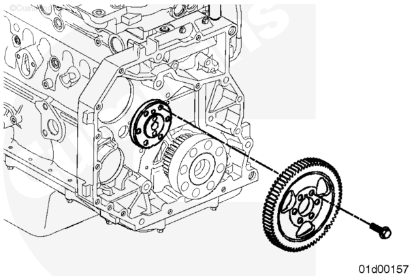
Remove the two capscrews securing the air compressor support (2).

If equipped with a hydraulic pump driven off of the air compressor, it may be necessary to remove and/or loosen some or all of the mounting fasteners. Refer to the OEM instructions.

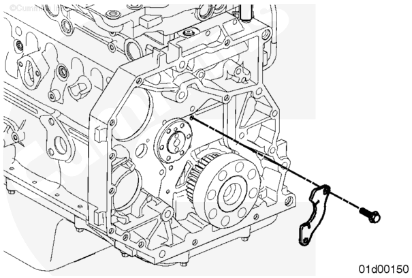
Remove the camshaft gear capscrews and remove the camshaft gear.

Refer to Procedure 001-012 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-012-tr.html).

Note : On engines equipped with an air compressor/accessory drive, it may be necessary to remove the air compressor/accessory drive to gain clearance to remove the camshaft gear.

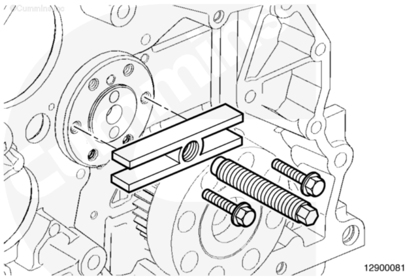


Remove the thrust plate capscrews and remove the thrust plate.



⚠ CAUTION ⚠

The camshaft will drop once the camshaft clears the last bushing if not supported. This can cause damage to the camshaft journal or, if equipped, the camshaft speed indicator ring.

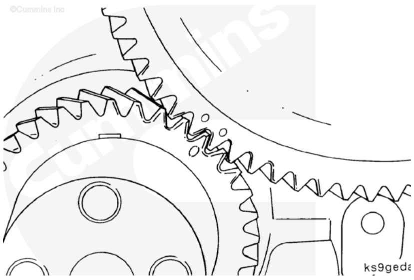


Use a gear puller, service tool part number ST647, or equivalent, to attach to the end of the camshaft where the camshaft gear mounts, to act as a handle. This will give proper leverage and make it easier to remove the camshaft.

Slide the camshaft out of the bore using the installed gear puller.

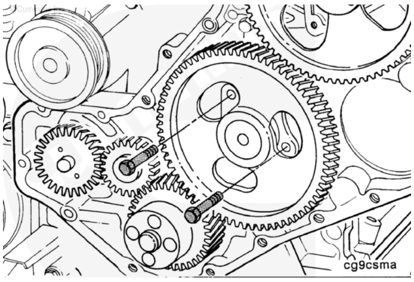
Front Gear Train

Use barring tool, Part Number 3824591, to rotate the crankshaft to align the crankshaft to the camshaft gear timing marks.



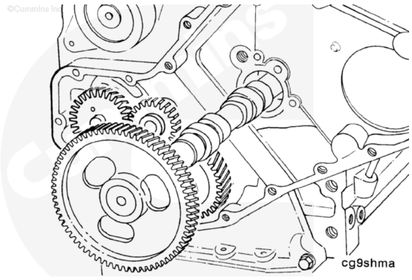
Remove the capscrews from the thrust plate.

Remove the thrust plate.



Remove the camshaft and camshaft gear as an assembly.

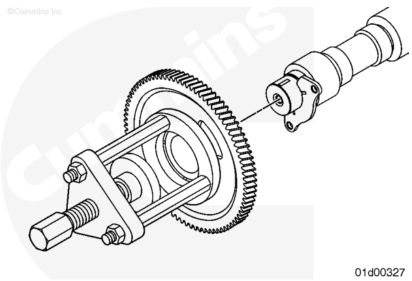
Note : Rotate the camshaft as it is being removed. Use extreme care to make sure that the bushings are not damaged during this process.



Disassemble

Front Gear Train

Remove the camshaft gear and locating key. Refer to Procedure 001-013 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-013.html).

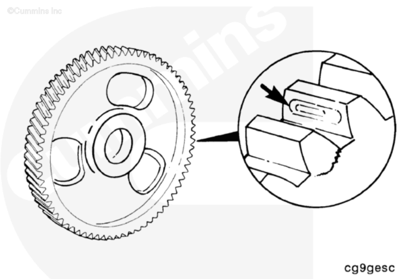


Clean and Inspect for Reuse

Inspect the camshaft gear.

For rear gear train engines, Refer to Procedure 001-012 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-012-tr.html).

For front gear train engines, Refer to Procedure 001-013 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-013.html).

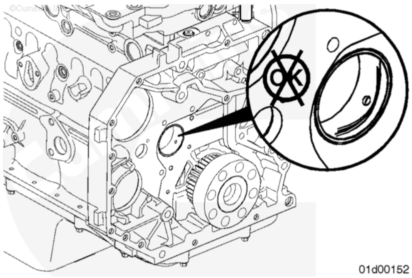


Inspect the camshaft bushing. Refer to Procedure 001-010 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-010-tr.html).

Note : Front gear train engines will have a front camshaft bushing. Rear gear train engines will have a rear camshaft bushing. Some engines may be equipped with both.

Only inspect the camshaft bushing that is on the same end of the engine from which the camshaft was removed.

Note : Inspection of the rest of the camshaft bushings and camshaft block bores is not necessary unless, during the inspection of the camshaft, damage was noted on the camshaft journals.



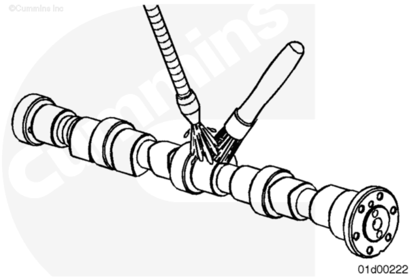
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

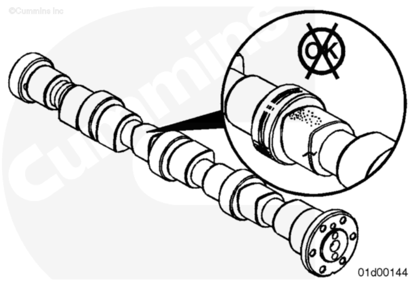
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the camshaft with solvents and dry with compressed air.

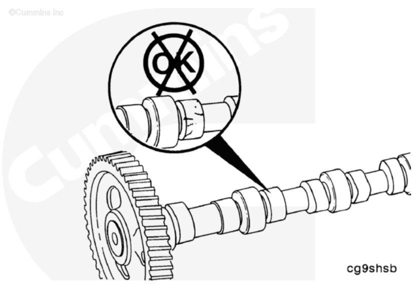


Inspect the valve lobes and bearing journals for cracking, pitting, or scoring.

Inspect the camshaft gear mounting surface on the camshaft to ensure the camshaft gear locating dowel pin is in place and **not** bent, sheared, or cracked.

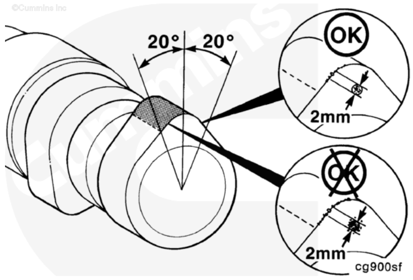


See Service Bulletin 3666475, Camshaft and Tappet Reuse Guidelines, for reuse guidelines for cast iron camshafts.



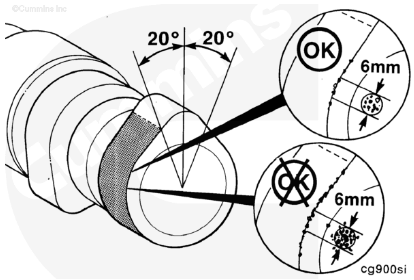
Edge Deterioration (breakdown) Criteria

The area of edge deterioration **must not** be greater than the equivalent area of a 2-mm [0.079-in] circle within ±20 degrees of the nose of the camshaft lobe.



Outside of the ±20 degrees of the nose of the camshaft lobe, the areas of edge deterioration **must not** be greater than the equivalent area of a 6 mm [0.236 in] circle.

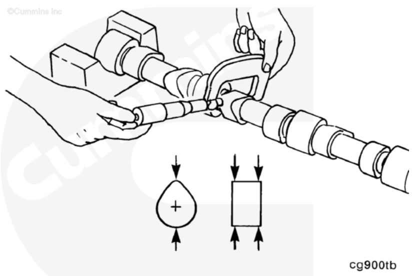
Note : If the camshaft shows any pitting or wear, remove and inspect the tappets before installing the camshaft. Refer to Procedure 004-015 in Section 4 (/qs3/pubsys2/xml/en/procedures/44/44-004-015.html). If a new camshaft is installed, new tappets and push rods must be installed also.



Measure

Measure the peak of the camshaft valve lobes.

4 Cylinder (Rear Gear Train) Diameter of Peak of Lobe			
	mm		in
Intake	46.132	MIN	1.8162
Exhaust	45.632	MIN	1.797



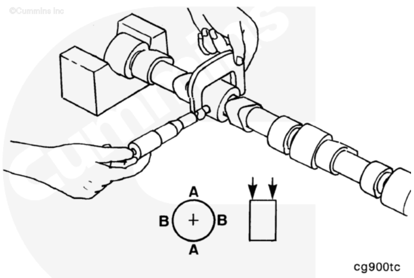
4 Cylinder (Front Gear Train) Engine Peak of Lobe Diameter by Camshaft Part Number		
Part Number	**Minimum Intake**	**Minimum Exhaust**
	mm [in]	**mm [in]**
3929039	47.392 [1.866]	47.122 [1.855]
3925582	47.392 [1.866]	47.122 [1.855]
3914638	47.392 [1.866]	47.122 [1.855]
3929885	47.803 [1.882]	46.609 [1.835]
3929038	47.803 [1.882]	47.122 [1.855]
3924574	47.803 [1.882]	47.122 [1.855]

4 Cylinder (Front Gear Train) Engine Peak of Lobe Diameter by Camshaft Part Number		
Part Number	**Minimum Intake**	**Minimum Exhaust**
	mm [in]	**mm [in]**
3931281	47.392 [1.866]	46.609 [1.835]
3930346	47.392 [1.866]	46.609 [1.835]

6 Cylinder (Front Gear Train) Engine Peak of Lobe Diameter by Camshaft Part Number		
Part Number	**Minimum Intake**	**Minimum Exhaust**
	mm [in]	mm [in]
3283179	47.803 [1.882]	46.609 [1.835]
3929734	47.803 [1.882]	47.122 [1.855]
3929040	47.803 [1.882]	47.122 [1.855]
3926671	47.803 [1.882]	47.122 [1.855]
3924109	42.811 [1.685]	47.122 [1.855]
3929041	47.392 [1.866]	47.122 [1.855]
3921953	47.392 [1.866]	47.122 [1.855]
3919608	47.392 [1.866]	47.122 [1.855]
3929042	47.392 [1.866]	47.122 [1.855]
3914639	47.392 [1.866]	47.122 [1.855]
3929886	47.803 [1.882]	46.609 [1.835]
3930378	47.392 [1.866]	46.609 [1.835]
3283179	47.803 [1.882]	46.609 [1.835]

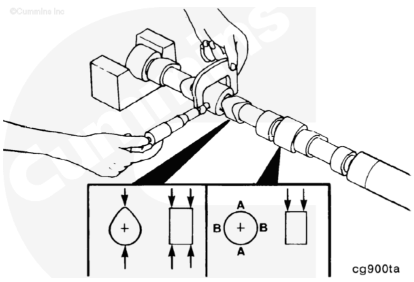
Measure the camshaft bearing journals.

Journal Diameter		
mm		in
53.962	MIN	2.1245
54.013	MAX	2.1265



Measure the fuel transfer pump lobe:

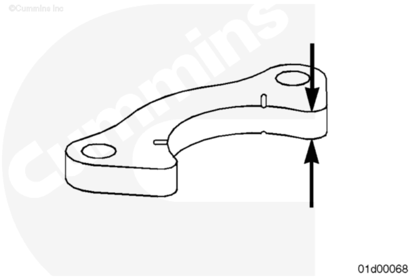
Fuel Transfer Pump Lobe Diameter		
mm		in
35.50	MIN	1.398
36.26	MAX	1.428



Measure the camshaft thrust plate thickness:

Rear Gear Train Camshaft Thrust Plate Thickness		
mm		in
5.25	MIN	0.207
5.35	MAX	0.211

Front Gear Train Camshaft Thrust Plate Thickness		
mm		in
9.40	MIN	0.370
9.60	MAX	0.378



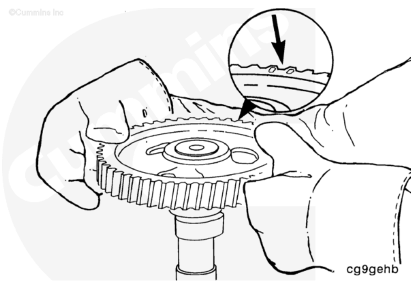
If the camshaft thrust plate is out of specification, replace the thrust plate.

Note : Front Gear Train and Rear Gear Train engines do not use the same camshaft thrust plate. The camshaft thrust plate thickness can also be verified by checking camshaft end play during installation.

Assemble

Front Gear Train

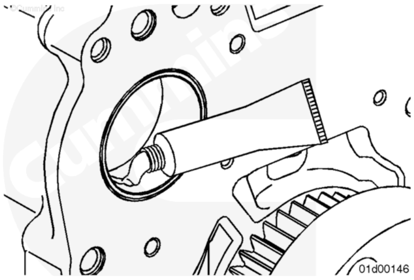
Install the camshaft gear locating key and camshaft gear. Refer to Procedure 001-013 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-013.html).



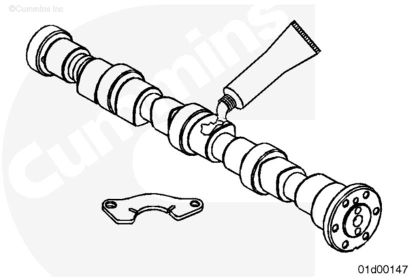
Install

Rear Gear Train

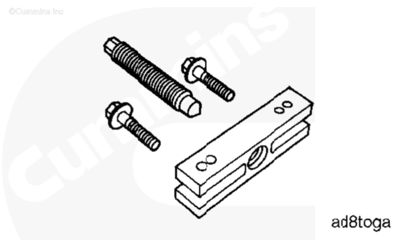
Apply assembly lubricant, Part Number 3163087, to the rear camshaft bore.



Lubricate the camshaft lobes, journals, and thrust washer with assembly lubricant, Part Number 3163087.

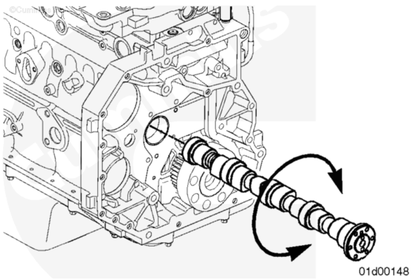


Use a gear puller, service tool Part Number ST647, or equivalent, to attach to the end of the camshaft where the camshaft gear mounts, to act as a handle. This will give proper leverage and ease installing the camshaft.



⚠ CAUTION ⚠

Do not force the camshaft into the camshaft bore as damage to the camshaft bushing can result.

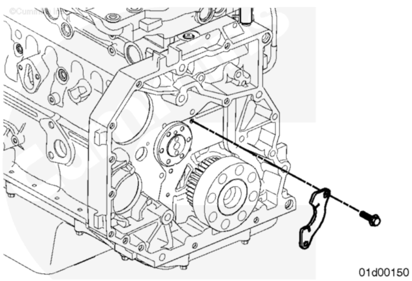


Install the camshaft. While pushing in slightly, rotate the camshaft and carefully work the camshaft through the camshaft bushings. As each camshaft journal passes through a bushing, the camshaft will drop slightly and the camshaft lobes will catch on the bushings. Rotating the camshaft will free the lobe from the bushing and allow the camshaft to be installed.

Install the thrust plate.

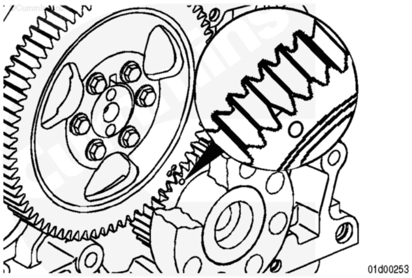
Install the thrust plate capscrews.

Torque Value: 24 n•m [212 in-lb]



⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the camshaft rotates freely.

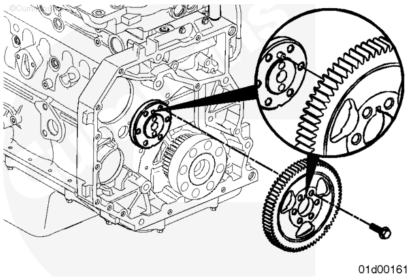


Note : The engine can have either a mark on the crankshaft gear or a chamfered tooth.

Align the timing marks on the camshaft gear with the timing marks on the crankshaft gear.

Note : If equipped with an air compressor, make sure to align the line that was scribed on the camshaft gear and air compressor gear during the camshaft gear removal step. Refer to Procedure 012-014 in section 12. (/qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html)If this was not done.

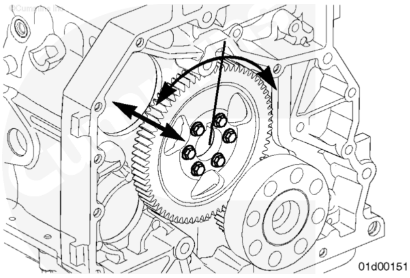
Rotate the camshaft so that the camshaft dowel pin aligns with the slot on the camshaft gear. Install the camshaft gear and capscrews.



Tighten the capscrews.

Torque Value: 36 n•m [27 ft-lb]

Use gauge, Part Number 3824564, and magnetic base, Part Number 3377399, to verify the camshaft has proper backlash and end clearance.

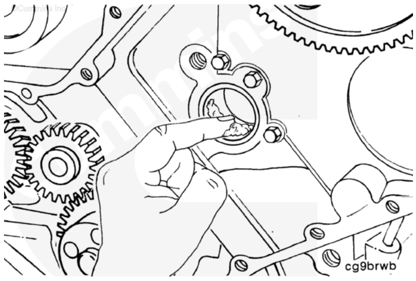


Camshaft End Play (A)		
mm		in
0.10	MIN	0.004
0.36	MAX	0.014

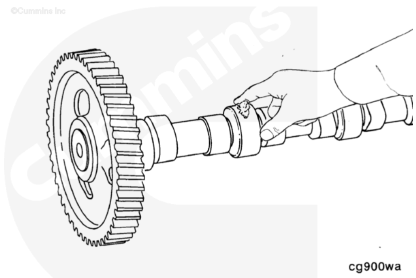
Camshaft Gear Backlash Limits (B)		
mm		in
0.076	MIN	0.003
0.280	MAX	0.011

Front Gear Train

Apply assembly lubricant, Part Number 3163087, to the front camshaft bore.

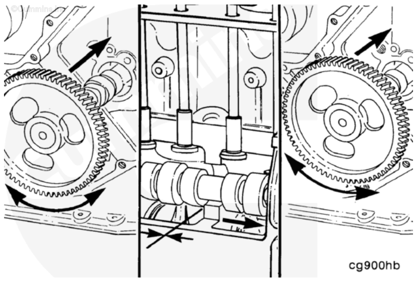


Lubricate the camshaft lobes, journals, and thrust plate with assembly lubricant, Part Number 3163087.



⚠ CAUTION ⚠

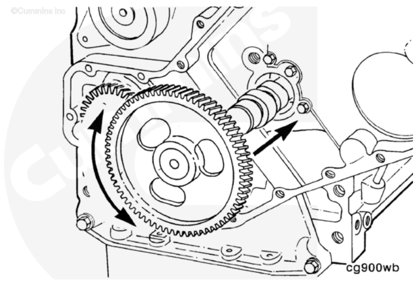
Do not try to force the camshaft into the camshaft bore as damage to the camshaft bushing can result.



Install the camshaft. While pushing in slightly, rotate the camshaft and carefully work the camshaft through the camshaft bushings. As each camshaft journal passes through a bushing, the camshaft will drop slightly and the camshaft lobes will catch on the bushings. Rotating the camshaft will free the lobe from the bushing and allow the camshaft to be installed.

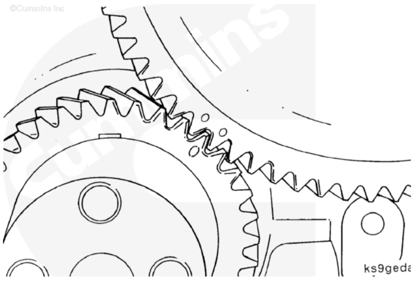
⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the camshaft rotates freely.



Before the camshaft gear engages the crankshaft gear, check the camshaft for ease of rotation. When installed properly, the camshaft **must** rotate freely.

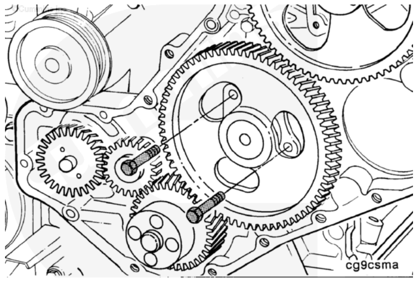
Align the timing marks as illustrated and finish installing the camshaft.



Install the thrust plate.

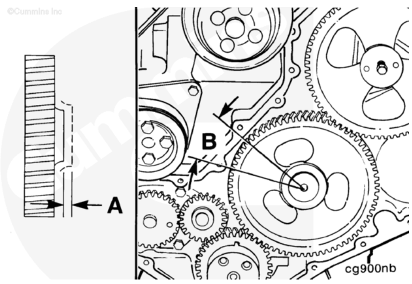
Install the thrust plate capscrews.

Torque Value: 24 n•m [18 ft-lb]



Use gauge, Part Number 3824564, and magnetic base, Part Number 3377399, to verify the camshaft has proper backlash and end clearance.

Camshaft End Play (A)		
mm		in
0.12	MIN	0.005
0.47	MAX	0.018
Camshaft Gear Backlash Limits (B)		
mm		in
0.076	MIN	0.003
0.280	MAX	0.011



Finishing Steps

Rear Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

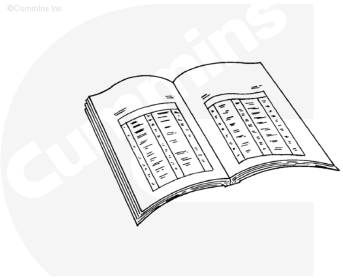
- Unlock the fuel pump. Refer to Procedure 005-014 in Section 5 (/qs3/pubsys2/xml/en/procedures/40/40-005-014.html).
- Lower the tappets. Refer to Procedure 004-015 in Section 4 (/qs3/pubsys2/xml/en/procedures/40/40-004-015-tr.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4 (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- Install the rocker levers. Refer to Procedure 003-008 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)
- Adjust the valve lash. Refer to Procedure 003-004 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Install the fuel lift pump. Refer to Procedure 005-045 in Section 5 (/qs3/pubsys2/xml/en/procedures/40/40-005-045.html).
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16 (/qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html)
- Install the flywheel. Refer to Procedure 016-005 in Section 16 (/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html)
- Install the flexplate. Refer to Procedure 016-004 in Section 16. (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Install the transmission and all related components, if equipped. Refer to the OEM service manual.
- Install the air compressor. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html)
- Connect the batteries
- Operate the engine and check for leaks.

Front Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the front gear cover. Refer to Procedure 001-031 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html)
- Install the vibration damper. Use the following procedure, if equipped with a viscous damper. Refer to Procedure 001-052 in Section 1 (/qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html).
- Use the following procedure, if equipped with a rubber damper. Refer to Procedure 016-004 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Release the tappets. Refer to Procedure 004-015 in Section 4 (/qs3/pubsys2/xml/en/procedures/40/40-004-015-tr.html)
- Install the push rods. Refer to Procedure 004-014 in section 4 (/qs3/pubsys2/xml/en/procedures/40/40-004-014-tr.html)
- Install the rocker levers. Refer to Procedure 003-008 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html)
- Adjust the valve lash. Refer to Procedure 003-004 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html)
- Install the fuel lift pump. Refer to Procedure 005-045 in Section 5 (/qs3/pubsys2/xml/en/procedures/40/40-005-045.html).
- Install the fan hub, if required. Refer to Procedure 008-039 in Section 8 (/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
- Install the drive belt. Refer to Procedure 008-002 in Section 8 (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html).
- Connect the batteries
- Operate the engine and check for leaks.

Last Modified: 27-Mar-2024